Vacuum-Weight Duality for Retained-State Defect Transfer
An End-to-End Development of Hypothesis H1

Research programme note

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Abstract

We develop Hypothesis H1 from a positive finite transfer amplitude through vacuum normalization, state-weighted defect transfer, connected-chain bounds, and the current Yang–Mills analytic frontier. The development corrects an important convention: the positive vacuum eigenfunction ψ enters the Doob transform, while the natural forward rare-state cost weight is $V = \psi^{-1}$. For a masked bad kernel this choice cancels the vacuum amplitude ratios exactly. Independently, the Perron vector of a bad subkernel minimizes its positive weighted ceiling. Applying this theorem to the package's four-channel star kernel proves that the previously selected weight is exactly optimal: its weighted ceiling is 0.002430277501841998, while its scalar row ceiling is one. The resulting degree-18 connected-cluster bound is strictly subcritical. These statements close H1 for finite positive kernels and conditional transfer amplitudes. A physical Yang–Mills theorem still requires construction of the normalized nonlinear retained-state kernel, a reflection-compatible eigenweight, and volume- and cutoff-uniform bad-row and root-moment bounds.

Contents

1 H1 and the convention problem **1**

2 Positive transfer amplitudes and vacuum normalization **2**

3 Optimality for a fixed defect kernel **3**

4 Exact four-channel retained-state certificate **3**

5 From one weighted row to connected clusters **4**

6 A positive two-sector transfer model **5**

6.1 Finite positivity is not uniformity 5

7 Exact normalized-kernel interface for Yang–Mills **5**

8 End-to-end audit **6**

1 H1 and the convention problem

The original intuitive statement was that the retained-state weight should be a coarse vacuum amplitude. That formulation merges three different objects.

(i) A positive transfer amplitude T has a principal right eigenfunction ψ, satisfying $T\psi = \lambda_0\psi$.

(ii) The vacuum Doob transform uses the ratio ψ_j/ψ_i to convert T into a normalized state-transition kernel.

(iii) A forward rare-state exploration uses a Lyapunov cost weight V. In the vacuum-derived construction the natural orientation is $V_i \propto \psi_i^{-1}$, not $V_i \propto \psi_i$.

There is also an independent optimization: for a fixed bad subkernel Q, its positive Perron vector minimizes

$$\sup_i \frac{(QV)_i}{V_i}. \tag{1}$$

This optimal defect weight need not be the reciprocal of the vacuum eigenfunction of a larger transfer operator.

The sharpened H1 is therefore:

> When retained-state dynamics is obtained by a Doob transform of a positive vacuum transfer amplitude, inverse vacuum amplitude supplies a canonical forward defect-cost weight. For the resulting bad subkernel, Perron optimization gives the best positive weighted ceiling. Identifying either vector with a physical Yang–Mills vacuum requires a reflection-positive temporal transfer interpretation.

This formulation separates physical time evolution, RG blocking, and an exploration kernel. A positive eigenvector of the latter two is not a physical vacuum merely because the same linear algebra appears.

2 Positive transfer amplitudes and vacuum normalization

Let $\Sigma = \{1, \dots, N\}$, and let T be an irreducible nonnegative matrix with principal eigenvalue $\lambda_0 > 0$, positive right eigenvector ψ, and positive left eigenvector φ:

$$T\psi = \lambda_0\psi, \qquad \varphi^{\mathsf{T}}T = \lambda_0\varphi^{\mathsf{T}}. \tag{2}$$

Normalize $\varphi^{\mathsf{T}}\psi = 1$.

Theorem 1 (Vacuum Doob transform)**.** *The matrix*

$$P_{ij} = \frac{T_{ij}\psi_j}{\lambda_0\psi_i} \tag{3}$$

is stochastic. The probability vector

$$\pi_i = \varphi_i\psi_i \tag{4}$$

is stationary. If T is symmetric, then $\pi_i = \psi_i^2/\sum_j \psi_j^2$, and P is reversible with respect to π.

Proof. The eigenvector equation gives $\sum_j P_{ij} = (T\psi)_i/(\lambda_0\psi_i) = 1$. Moreover,

$$\sum_i \pi_i P_{ij} = \sum_i \frac{\varphi_i T_{ij}\psi_j}{\lambda_0} = \varphi_j\psi_j = \pi_j.$$

For symmetric T, left and right eigenvectors coincide up to normalization, and $\pi_i P_{ij} = \psi_i T_{ij}\psi_j/\lambda_0 = \pi_j P_{ji}$. □

Let $M_{ij} \in [0,1]$ be a bad-edge mask and define the bad subkernel

$$Q_{ij} = M_{ij} P_{ij}. \tag{5}$$

Theorem 2 (Inverse-amplitude cancellation)**.** *Set $V_i = c/\psi_i$, with $c > 0$ chosen so that $\min_i V_i = 1$. Then*

$$\frac{(QV)_i}{V_i} = \frac{1}{\lambda_0} \sum_j M_{ij} T_{ij}. \tag{6}$$

Consequently,

$$QV \le q_{\mathrm{vac}} V, \qquad q_{\mathrm{vac}} = \frac{1}{\lambda_0} \max_i \sum_j M_{ij} T_{ij}. \tag{7}$$

Proof. Substitute the definitions:

$$\frac{1}{V_i} \sum_j M_{ij} \frac{T_{ij}\psi_j}{\lambda_0 \psi_i} V_j = \frac{1}{V_i} \sum_j M_{ij} \frac{T_{ij}\psi_j}{\lambda_0 \psi_i} \frac{c}{\psi_j} = \frac{1}{\lambda_0} \sum_j M_{ij} T_{ij}.$$

□

Writing $I_i = -\log \psi_i$, the forward weight is $V_i \propto e^{I_i}$. It records the cost of visiting a vacuum-suppressed boundary state. The inverse orientation is therefore a substantive part of H1.

3 Optimality for a fixed defect kernel

The vacuum-derived weight is canonical when a physical T is available. For a fixed bad kernel Q, a separate variational statement identifies the best possible positive diagonal weight.

Theorem 3 (Collatz–Wielandt defect optimum)**.** *Let Q be a finite irreducible nonnegative matrix. Then*

$$\rho_{\mathrm{PF}}(Q) = \inf_{V>0} \max_i \frac{(QV)_i}{V_i} = \sup_{V>0} \min_i \frac{(QV)_i}{V_i}. \tag{8}$$

The infimum is achieved by a positive Perron vector V_{PF}, for which $QV_{\mathrm{PF}} = \rho_{\mathrm{PF}}(Q) V_{\mathrm{PF}}$.

Thus no positive weighting can produce a ceiling below the spectral radius of the declared bad subkernel. This is an exact obstruction as well as an optimization theorem.

Warning 1. *Computing V_{PF} from a finite kernel does not prove a uniform Yang–Mills estimate. The kernel must first be the complete normalized physical conditional kernel. Its state space, positivity, boundary dependence, topology, and cutoff dependence cannot be inferred from Perron theory.*

For an exploration or RG kernel, V_{PF} is a Lyapunov eigenweight. Calling it a vacuum wavefunction is justified only after an independent theorem identifies that kernel with reflection-positive temporal transfer.

4 Exact four-channel retained-state certificate

Let the regular state be 0, and let states $1, \dots, k$ denote charged defect channels. For $0 < q < 1$, define

$$Q_{0a} = \frac{q^2}{k}, \qquad Q_{a0} = 1, qquad Q_{00} = Q_{ab} = 0 \quad (a, b \ge 1). \tag{9}$$

The scalar row ceiling is one because return from a charged state is forced.

Proposition 1 (Exact star eigenweight)**.** *For*

$$V_0 = 1, \qquad V_a = q^{-1} \quad (1 \le a \le k), \tag{10}$$

one has

$$QV = qV. \tag{11}$$

Hence $\rho_{\mathrm{PF}}(Q) = q$, and the declared weight is the optimal positive weight.

Proof. The regular row is

$$(QV)_0 = \sum_{a=1}^{k} \frac{q^2}{k} \frac{1}{q} = q = qV_0.$$

Every charged row is $(QV)_a = V_0 = 1 = qV_a$. Positivity and Theorem 3 gives optimality. □

For the package calibration,

$$q = 0.002430277501841998, \qquad k = 4, \tag{12}$$

with channels center, continuous, cut/orbit, and integer topology. The charged-state weights are

$$q^{-1} = 411.4756439304\ldots. \tag{13}$$

The entry mass per channel is $q^2/4 = 1.4765621839898456 \times 10^{-6}$. The exact weighted ceiling is q, while the scalar ceiling is one. The numerical eigen-residual in the reproducibility certificate is below 9×10^{-19}.

This upgrades the earlier finite calibration: the chosen weight was not only feasible but Perron-optimal for its declared multichannel star kernel.

5 From one weighted row to connected clusters

Let $Q_1, \ldots, Q_n$ be positive bad-extension kernels on a common retained state space and suppose

$$Q_j V \le qV \tag{14}$$

for every exploration history. If the initial state law ν satisfies $\nu(V) \le M$, positivity gives

$$\nu Q_1 \cdots Q_n \mathbf{1} \le \nu Q_1 \cdots Q_n V \le M q^n. \tag{15}$$

The terminal inequality uses $V \ge 1$.

On a graph of maximum degree Δ, use the depth-first upper bound $\Delta^{2(n-1)}$ for rooted connected sets of size n. If

$$e^{\alpha} \Delta^2 q < 1, \tag{16}$$

then

$$\sum_{\substack{X \ni o \\ X \text{ connected}}} e^{\alpha |X|} \mathbb{P}(B_X) \le \frac{M e^{\alpha} q}{1 - e^{\alpha} \Delta^2 q}. \tag{17}$$

At $\Delta = 18$, $\alpha = 0.1$, and the calibrated q,

$$e^{0.1} 18^2 q = 0.8702225337961371, \tag{18}$$

$$q_{\mathrm{crit}} = \frac{e^{-0.1}}{18^2} = 0.0027927080803578997, \tag{19}$$

$$\frac{e^{0.1} q}{1 - e^{0.1} 18^2 q} = 0.020695981332153383. \tag{20}$$

The margin is strict. Equality at $q = q_{\mathrm{crit}}$ is rejected.

The executable certificate enumerates the exact four-channel bad-chain mass through length twelve. Every weighted bound holds; even lengths saturate in the alternating star model, whereas odd-length bounds retain the entry-cost slack.

6 A positive two-sector transfer model

To test the vacuum interpretation rather than only Perron optimization, take

$$T_\varepsilon = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 2\varepsilon^2 \end{pmatrix}, \qquad 0 < \varepsilon < 2^{-1/2}. \tag{21}$$

It is symmetric, entrywise nonnegative, and positive definite because its determinant is ε^2. Declare a bad extension to be a transition whose next state is sector one.

At $\varepsilon = 0.002430277501841998$, the certificate gives

$$\lambda_0 = 1.0000059062836195, \tag{22}$$

$$\psi = (1,\ 0.002430291855665421), \tag{23}$$

$$V = \psi^{-1} = (1,\ 411.4732136672519). \tag{24}$$

The Doob row-sum error is below 2.3×10^{-16}, the stationary and detailed-balance checks pass at the same scale, and

$$\max_i \frac{(QV)_i}{V_i} = 0.002430263148018576. \tag{25}$$

The identity (6) holds with error below 4.4×10^{-19}.

Using ψ instead of ψ^{-1} in the forward cost inequality does not yield the cancellation identity; the recorded error is 2.4302488×10^{-3}. This does not say that amplitude weighting is never useful. It says that the two orientations answer different questions and cannot be interchanged silently.

6.1 Finite positivity is not uniformity

Let a sequence of finite volumes have a rare-sector amplitude $\psi_L = e^{-L}$. Every finite-volume vacuum is positive, but the inverse cost weight grows as e^L. At $L = 20$ it exceeds 4.85×10^8. Perron positivity alone therefore supplies neither a volume-uniform Harnack bound nor a uniform root moment. Those are separate analytic obligations.

7 Exact normalized-kernel interface for Yang–Mills

For a retained boundary state s, compact conditional fiber F_s, positive fiber weight $w_s(f)$, next-state map $\tau(s, f)$, and bad region B_s, define

$$Z(s) = \int_{F_s} w_s(f)\, df, \tag{26}$$

$$Q(s, A) = \frac{1}{Z(s)} \int_{B_s} \mathbf{1}_{\{\tau(s,f)\in A\}} w_s(f)\, df. \tag{27}$$

For any positive V, the exact weighted bad row is

$$q_V(s) = \frac{1}{Z(s)V(s)} \int_{B_s} w_s(f) V(\tau(s, f))\, df. \tag{28}$$

No action-only estimate can replace the normalizer $Z(s)$.

H1 proposes to obtain or approximate V from a positive principal eigenfunction. There are two admissible routes.

Temporal-vacuum route. Construct a reflection-positive temporal transfer operator $T_{a,L}$, prove a strictly positive vacuum eigenfunction $\psi_{a,L}$, and show that the retained-state conditional kernel is its Doob transform or a controlled approximation. Then use $V_{a,L} = \psi_{a,L}^{-1}$.

Exploration-Perron route. Construct the complete normalized bad subkernel $Q_{a,L}$ directly and find a positive Lyapunov eigenweight. This can prove defect suppression, but the eigenweight is not called a physical vacuum without a temporal-transfer identification.

Both routes must prove, uniformly over finite volume, cutoff, and every positive-mass retained history,

$$Q_{a,L}V_{a,L} \le qV_{a,L}, \qquad q < q_{\mathrm{crit}}, \tag{29}$$

$$\nu_{a,L}(V_{a,L}) \le M < \infty. \tag{30}$$

The state must retain continuous collar, center, cut/orbit, and compositionally sufficient integer topology data. Gauge covariance, reflection compatibility, chart complements, partition normalizers, and cross-channel persistence remain explicit proof obligations.

The first open implication is therefore not Perron theory. It is the construction and uniform analytic control of the physical nonlinear kernel appearing in (28).

8 End-to-end audit

Obligation	Status	Evidence or blocker
Finite positive-transfer/Doob theorem	proved	Exact algebra and executable two-sector certificate.
Inverse-vacuum-amplitude forward weight	proved	Exact cancellation identity for every finite masked transfer matrix.
Perron-optimal defect weight	proved	Collatz–Wielandt theorem; exact four-channel eigenweight.
Bad-chain and rooted-cluster iteration	proved	Twelve-step enumeration and strict degree-18 threshold.
Cross-stage parameter consistency	proved	Stages 19, 24, and 25 agree on q and branching constants.
Complete normalized nonlinear SU(2) kernel	open	Current package has only restricted/reference rows and conditional interfaces.
Physical vacuum interpretation	open	Requires a reflection-positive temporal transfer operator and identification with the retained kernel.
Uniform inverse-weight/root moment	open	Finite positivity does not prevent exponential volume growth.

Obligation	Status	Evidence or blocker
Full multichannel topology and chart complement	open	Integer sector weights and boundary-uniform complement bounds are missing.
One-scale basin entry and repeated RG	open	Depend on the preceding physical kernel inequality.
Continuum construction and Hamiltonian gap	open	OS axioms, density, uniform temporal rigidity, and theory identity remain unproved.

The strongest defensible level is C2: finite positive-transfer and state-weighted-kernel theorems, plus a conditional interface to the physical measure. H1 is mathematically developed end to end as a mechanism. Its central Yang–Mills premise is not established.

Vacuum amplitude normalizes the dynamics; inverse vacuum amplitude prices rare retained states; the bad-kernel Perron vector optimizes the price.